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UNIVERSITI
TEKNOLOGI
MARA

COLLEGE OF COMPUTING, INFORMATICS AND MATHEMATICS

BACHELOR OF INFORMATION SYSTEM (HONS.)

CS240

CSC584 ENTERPRISE PROGRAMMING

GROUP: CDCS2534A

PREPARED BY

NAME	STUDENT ID	GROUP
MUHAMMAD IRFAN UZAIR BIN MAD HUSSAIN	2025494272	NBCS2404A
MUHAMMAD HABIL BIN MOHAMAD IQBAL	2023283064	NBCS2404A
MUHAMMAD EMIR SHAH BIN MOHAMED	2023282688	NBCS2404A
MUHAMMAD LUQMAN HAKIM BIN KHAIRUL SALLEH	2025616502	NBCS2404A

LECTURER NAME : MUHAMMAD RIDHWAN BIN MOHAMAD RAZALI

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1. INTRODUCTION

The UITM Election Management System is a web based application designed to simplify and automate the process of conducting campus elections at Uitm. This system enables administrators to efficiently manage student registrations, candidate nominations, election processes, and voting operations, while allowing students to securely cast their votes online. The system ensures a transparent, fair, and efficient election process for both administrators and student voters.

2. PROBLEM STATEMENT

Traditional campus election systems face significant challenges in managing student participation, candidate registration, and vote counting efficiently. Existing manual or semi automated systems lack features such as secure student authentication, real time vote tracking, comprehensive CRUD (Create, Read, Update, Delete) operations for election management, and clear dashboards for monitoring election progress and results. This leads to time consuming processes, potential errors in vote counting, and limited accessibility for students.

3. OBJECTIVE

To design and develop a comprehensive digital solution for managing campus elections, including student registration, candidate management, election administration, and secure online voting, while providing real time monitoring and result analysis capabilities.

4. DEMONSTRATION

This demonstration showcases the UITM Election Management System, designed to streamline the process of conducting campus elections through secure student authentication, efficient data management (CRUD operations), and comprehensive dashboards for monitoring elections, candidates, and voting activities. The system enables administrators to manage students, candidates, and elections seamlessly while providing students with a secure platform to cast their votes and view real time results.

4.1. STUDENT LOGIN PAGE

The screenshot displays the Student Login page of the UITM Election System. The page has a dark blue header with the text "UITM Election System" on the left and navigation links "Home", "About", "Contact", and "Register" on the right. The main content area is light gray and contains a white login box. On the left side of this box, there is a blue square with the text "UITM Election System" and "Secure Online Voting Platform". The right side of the box is titled "Login to Your Account" and contains two tabs: "Student Login" (active) and "Admin Login". Below the tabs are two input fields: "Student ID:" with a placeholder "Enter student ID" and "Password:" with a placeholder "Enter password". Below these fields are two buttons: a blue "Login as Student" button and a light blue "New Student? Click to Register" button. The footer is dark blue and contains two columns of text. The left column is titled "UITM Election System" and includes the text "Secure online voting platform for campus elections" and "Managed by University Technology MARA". The right column is titled "Contact Information" and includes "Email: election@uitm.edu.my", "Phone: 03-5544 2000", and "Location: Shah Alam Campus". At the bottom center, there is a copyright notice: "© 2024 UITM Election System. CSC584 Group Project."

FIGURE 1 STUDENT LOGIN PAGE

On this page, students can log in using their registered Student ID and password. Only authenticated students can access the voting dashboard and participate in ongoing elections. The system validates user credentials against the database and establishes a secure session upon successful login.

4.2. ADMIN LOGIN PAGE

UITM Election System

Home About Contact Register

UITM Election System
Secure Online Voting Platform

Login to Your Account

Student Login Admin Login

Username:
Enter admin username

Password:
Enter admin password

Login as Admin

New Student? Click to Register

UITM Election System
Secure online voting platform for campus elections
Managed by University Technology MARA

Contact Information
Email: election@uitm.edu.my
Phone: 03-5544 2000
Location: Shah Alam Campus

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FIGURE 2 ADMIN LOGIN PAGE

On this page, administrators can log in using their registered username and password. Only authenticated administrators can access the admin dashboard and manage system functionalities including student records, candidate nominations, election settings, and vote monitoring. The system validates admin credentials against the database and establishes a secure session upon successful login.

4.3. REGISTER PAGE

The screenshot displays the 'UITM Election System' web interface. At the top, a dark blue header contains the site name on the left and navigation links (Home, About, Contact, Register) on the right. The main content area is light gray and features a white registration form titled 'Student Registration'. Below the title is a subtitle: 'Create your account to vote in UITM elections'. The form includes input fields for 'Student ID' and 'Full Name', a single-line field for 'Email', and two-line fields for 'Password' and 'Confirm Password'. A dropdown menu for 'Faculty' is set to 'Faculty of Computer Science & IT'. At the bottom of the form are a blue 'Register Account' button and a 'Back to Login Page' link. The footer is a dark blue bar with the text 'UITM Election System', 'University Technology MARA', and 'Secure Campus Voting Platform'.

UITM Election System

Home About Contact Register

Student Registration

Create your account to vote in UITM elections

Student ID: Full Name:

Email:

Password: Confirm Password:

Faculty:
Faculty of Computer Science & IT

Register Account

[Back to Login Page](#)

UITM Election System
University Technology MARA
Secure Campus Voting Platform

FIGURE 3 REGISTER PAGE

On this page, new students can create an account by inserting their student ID, full name, email, password, and select faculty. Upon successful registration, the system stores their information in the database and redirects them to the login page. This page ensures that only registered students can access the voting system.

4.4. STUDENT DASHBOARD

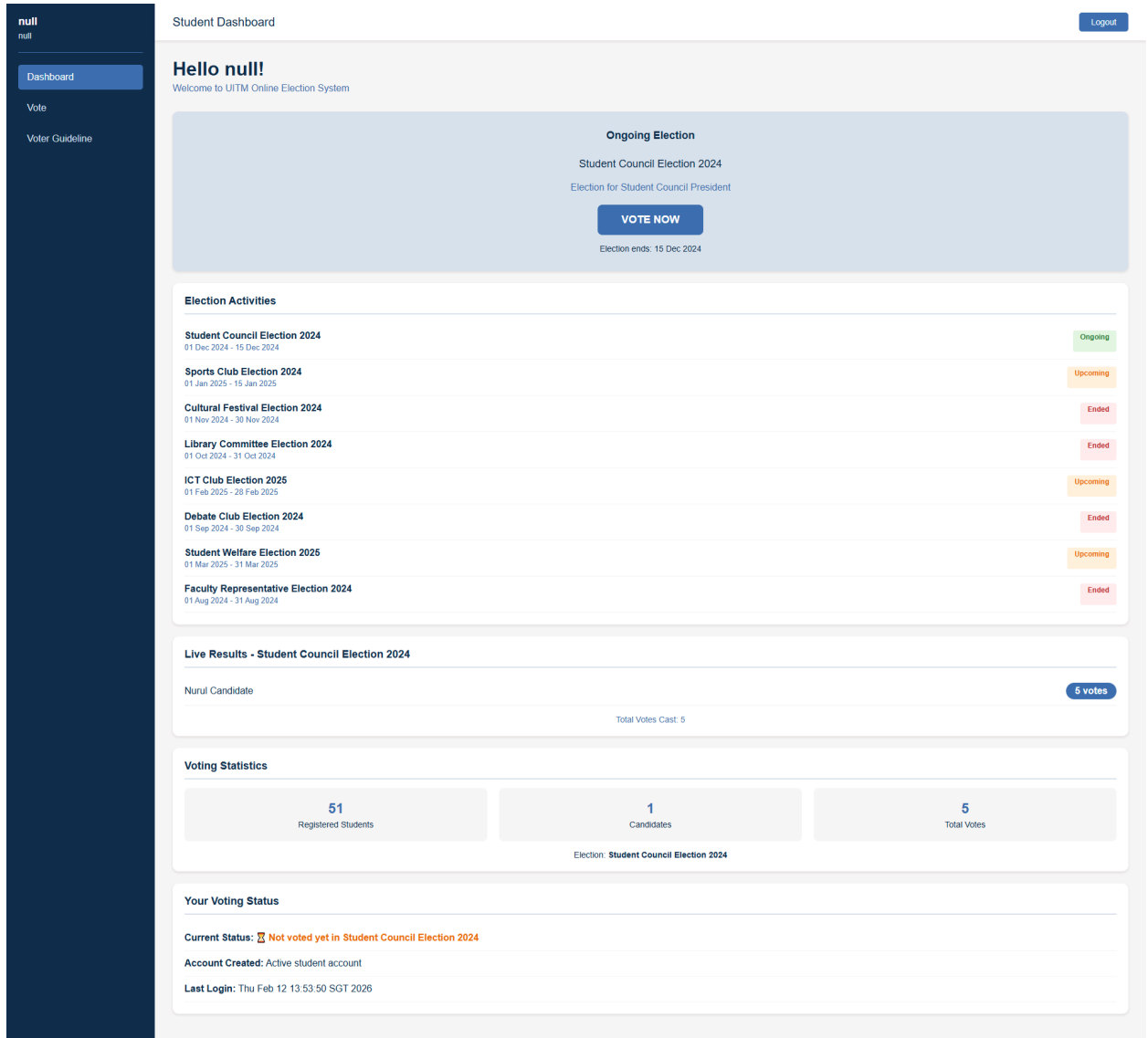


FIGURE 4 STUDENT DASHBOARD

On this page, students can view ongoing elections, candidate information, and their voting status. The dashboard displays a welcome message, election details, and a "VOTE NOW" button for active elections. Students who have already voted will see a confirmation message instead. The page also shows live results, voting statistics, and quick access to voting guidelines, providing a centralized hub for student participation.

4.5. VOTE PAGE

The screenshot displays the 'Voting Page' of a student election system. On the left is a dark blue sidebar with the user's name 'null' at the top, followed by navigation links: 'Dashboard', 'Vote' (highlighted in blue), and 'Voter Guideline'. The main content area is titled 'Voting Page' and features a 'Logout' button in the top right corner. It contains four election cards:

- Student Council Election 2024**: Election for Student Council President • Status: **ONGOING** • Ends: 2024-12-15. It features a candidate card for 'Nurul Candidate' (President) with the manifesto 'Promote cultural diversity and inclusion'. The 'VOTE' button is active (blue), while the 'Details' button is disabled (grey).
- Sports Club Election 2024**: Election for Sports Club Committee • Status: **UPCOMING** • Starts: 2025-01-01. A warning message states: '⚠ This election has not started yet. Voting will open on 2025-01-01.' The candidate card for 'Siti Candidate' (Sports Secretary) with the manifesto 'Improve sports facilities and equipment' has both 'VOTE' and 'Details' buttons disabled (grey).
- ICT Club Election 2025**: Election for ICT Club President • Status: **UPCOMING** • Starts: 2025-02-01. A warning message states: '⚠ This election has not started yet. Voting will open on 2025-02-01.' No candidate card is visible.
- Student Welfare Election 2025**: Election for Student Welfare Committee • Status: **UPCOMING** • Starts: 2025-03-01. A warning message states: '⚠ This election has not started yet. Voting will open on 2025-03-01.' No candidate card is visible.

FIGURE 5 VOTE PAGE

On this page, students can cast their votes for candidates in ongoing elections. The page displays all candidates grouped by position, along with their manifestos and details. Students must confirm their selection before submitting. The system enforces one vote per election rules and prevents duplicate voting.

4.6. VOTING GUIDLELINE PAGE

UITM Election Guidelines

Please read and follow these guidelines for a fair and transparent election process.

General Rules

- 1 Only registered UITM students are eligible to vote.
- 2 Each student can vote only ONCE per election.
- 3 Voting is confidential and cannot be changed once submitted.
- 4 Do not share your login credentials with anyone.
- 5 Voting period: 8:00 AM to 6:00 PM on election day.

Voting Process

- 1 Login to the election system using your student credentials.
- 2 Select the election you wish to vote in.
- 3 Review all candidate information before voting.
- 4 Select your preferred candidate and confirm your vote.
- 5 Wait for confirmation message before closing the window.

Prohibited Actions

- 1 Do not attempt to vote multiple times.
- 2 Do not coerce or influence other voters.
- 3 Do not use someone else's account to vote.
- 4 Do not share your voting choice publicly.

⚠ IMPORTANT WARNING

Any violation of these guidelines will result in:

- Immediate disqualification of your vote
- Disciplinary action by the university
- Possible suspension of voting privileges

IMPORTANT NOTE

If you encounter any technical issues during voting:

1. Contact Election Technical Support immediately
2. Do NOT attempt to vote from another device
3. Report the issue with your student ID and time of error

Last updated: December 2024
UITM Election Committee

FIGURE 6 VOTING GUIDLELINE PAGE

On this page, students can read the rules and steps for voting, including eligibility, how to vote, and prohibited actions. This helps ensure fair and transparent elections.

4.7. ADMINISTRATOR DASHBOARD

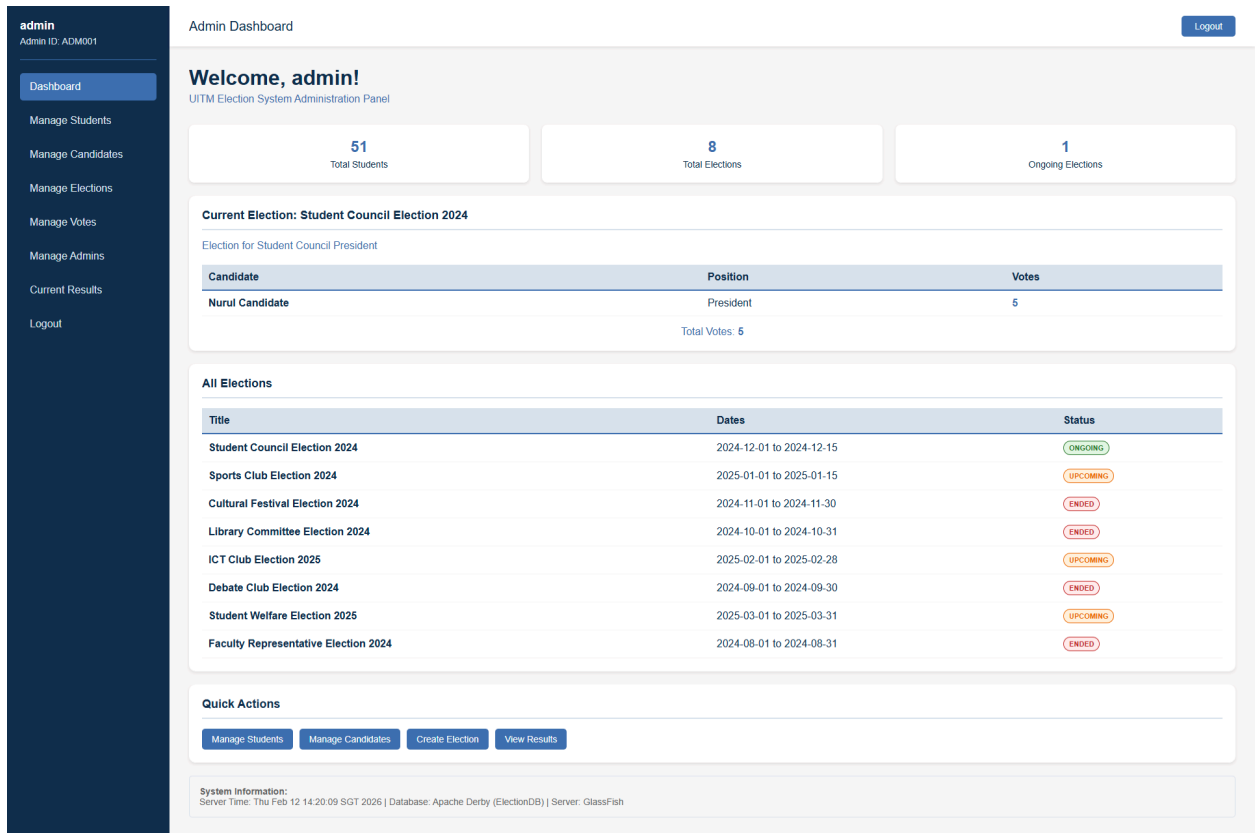


FIGURE 7 ADMINISTRATOR DASHBOARD

After login, administrators are directed to the Admin Dashboard, which serves as the central command center for the UITM Election System. The dashboard provides a comprehensive overview of key system statistics, including:

- **Total Registered Students:** The total number of student accounts registered in the system
- **Total Elections:** All elections created, regardless of status
- **Ongoing Elections:** Active elections currently accepting votes

The dashboard also features a quick access navigation panel that allows administrators to efficiently manage tables of the system:

1. Student management
2. Candidate management
3. Election management

4. Vote management
5. Admin account management
6. Election management

This centralized hub ensures that administrators can seamlessly oversee and control all election related activities from one intuitive interface.

4.8. MANAGE STUDENTS PAGE

The screenshot displays the 'Manage Students' interface. On the left is a dark sidebar with the user 'admin' (ID: ADM001) and navigation options: Dashboard, Manage Students (active), Manage Candidates, Manage Elections, Manage Votes, Manage Admins, Current Results, and Logout. The main content area has a 'Manage Students' header with a search bar and an '+ Add New Student' button. Below is a table titled 'Students List (51 students)'.

Student ID	Name	Email	Faculty	Voted	Candidate Status	Actions
2023	emir shah	gilaxe940@bulatox.com	null	Voted	voter	Edit Reset Password Delete
C2001	Ahmad Candidate	ahmad.c@uitm.edu.my	FCSIT	Not Voted	candidate	Edit Reset Password Delete
C2002	Siti Candidate	siti.c@uitm.edu.my	FKE	Not Voted	candidate	Edit Reset Password Delete
C2003	Hafiz Candidate	hafiz.c@uitm.edu.my	FBM	Not Voted	candidate	Edit Reset Password Delete
C2004	Nurul Candidate	nurul.c@uitm.edu.my	FSSH	Not Voted	candidate	Edit Reset Password Delete
C2005	John Candidate	john.c@uitm.edu.my	FSPU	Not Voted	candidate	Edit Reset Password Delete
S1001	Ahmad bin Ali	ahmad@uitm.edu.my	FCSIT	Voted	voter	Edit Reset Password Delete
S1002	Siti Sarah binti Kamal	siti@uitm.edu.my	FKE	Voted	voter	Edit Reset Password Delete

FIGURE 8 MANAGE STUDENTS PAGE

The Student Management page provides administrators with complete control over student voter accounts. This interface enables full CRUD (Create, Read, Update, Delete) operations, allowing administrators to:

- View: Display a comprehensive table of all registered students with their complete details
- Add: Register new student accounts by entering Student ID, name, email, password, and faculty
- Edit: Modify existing student information including name, email, faculty, and candidate status
- Search: Quickly filter and locate specific students using keywords or Student ID
- Delete: Remove student accounts from the system (with safety checks for candidates)

4.9. MANAGE CANDIDATES PAGE

admin
Admin ID: ADM001

Dashboard

Manage Students

Manage Candidates

Manage Elections

Manage Votes

Current Results

Logout

Manage Candidates

Search candidates...

Search

+ Add New Candidate

Candidates List (8 candidates)

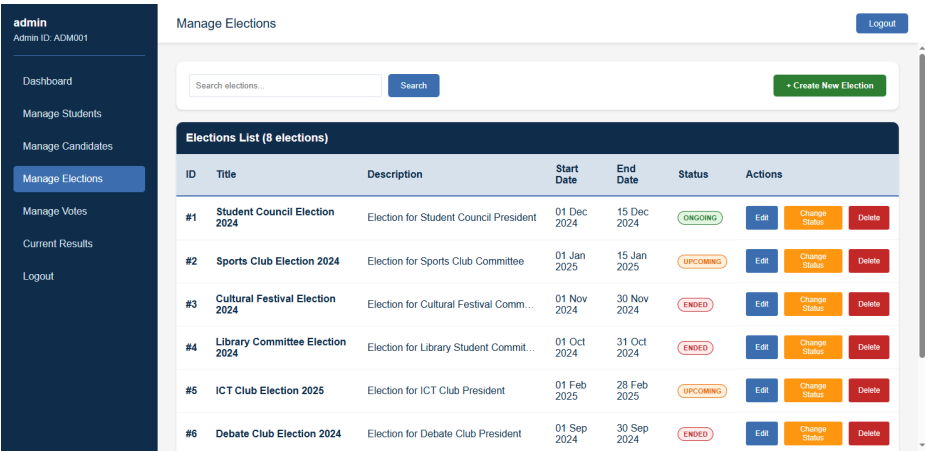
Candidate ID	Candidate Name	Student ID	Position	Election	Manifesto	Votes	Actions
4	Nurul Candidate	C2004	President	Student Council Election 2024	Promote cultural dive...	5	<a>Edit <a>Delete
6	Siti Candidate	C2002	Sports Secretary	Sports Club Election 2024	Improve sports faciliti...	4	<a>Edit <a>Delete
7	Hafiz Candidate	C2003	Festival Director	Cultural Festival Election 2024	Organize diverse cult...	8	<a>Edit <a>Delete
8	Nurul Candidate	C2004	Logistics Head	Cultural Festival Election 2024	Manage event logisti...	5	<a>Edit <a>Delete
9	John Candidate	C2005	Publicity Head	Cultural Festival Election 2024	Promote festival to all...	5	<a>Edit <a>Delete
11	Hafiz Candidate	C2003	Technology Officer	Library Committee Election 2024	Upgrade library comp...	0	<a>Edit <a>Delete

FIGURE 9 MANAGE CANDIDATES PAGE

The Candidate Management page enables administrators to manage all candidate registrations across different elections. This interface provides full control over the candidate nomination process with the following capabilities:

- **View:** Display a complete list of all candidates with their detailed information in a sortable table format
- **Add:** Register new candidates by linking existing student accounts to specific elections
- **Edit:** Modify candidate details including candidate name, position, and manifesto
- **Delete :** Delete candidate registrations from the system when necessary

4.10. MANAGE ELECTION PAGE



The screenshot shows the 'Manage Elections' page. On the left is a dark blue sidebar with the user 'admin' (ID: ADM001) and a menu including Dashboard, Manage Students, Manage Candidates, Manage Elections (highlighted), Manage Votes, Current Results, and Logout. The main content area has a 'Manage Elections' header with a search bar and a '+ Create New Election' button. Below is a table titled 'Elections List (8 elections)' with 8 rows. Each row contains an ID, Title, Description, Start Date, End Date, Status (with a color-coded badge), and a set of three action buttons: Edit, Change Status, and Delete.

ID	Title	Description	Start Date	End Date	Status	Actions
#1	Student Council Election 2024	Election for Student Council President	01 Dec 2024	15 Dec 2024	ONGOING	Edit Change Status Delete
#2	Sports Club Election 2024	Election for Sports Club Committee	01 Jan 2025	15 Jan 2025	UPCOMING	Edit Change Status Delete
#3	Cultural Festival Election 2024	Election for Cultural Festival Comm...	01 Nov 2024	30 Nov 2024	ENDED	Edit Change Status Delete
#4	Library Committee Election 2024	Election for Library Student Commit...	01 Oct 2024	31 Oct 2024	ENDED	Edit Change Status Delete
#5	ICT Club Election 2025	Election for ICT Club President	01 Feb 2025	28 Feb 2025	UPCOMING	Edit Change Status Delete
#6	Debate Club Election 2024	Election for Debate Club President	01 Sep 2024	30 Sep 2024	ENDED	Edit Change Status Delete

FIGURE 10 MANAGE ELECTION PAGE

The Election Management page provides administrators with complete control over the entire election lifecycle. This interface enables full CRUD (Create, Read, Update, Delete) operations, allowing administrators to:

1. **Create:** Schedule new elections by entering title, description, start date, end date, and initial status (Upcoming/Ongoing/Ended). Date validation ensures the end date occurs after the start date.
2. **View:** Display a comprehensive table of all elections with complete details including:
 - Election ID and Title
 - Description
 - Start Date and End Date
 - Current Status with color coded badges (● Ongoing, ● Upcoming, ● Ended)
3. **Edit:** Modify existing election information including title, description, dates, and status as election requirements change.
4. **Delete:** Remove elections from the system with builtin safety checks that prevent deletion when candidates or votes are already in.
5. **Search:** Quickly filter and locate specific elections using keywords, title, date range, or status.

4.11. MANAGE VOTES PAGE

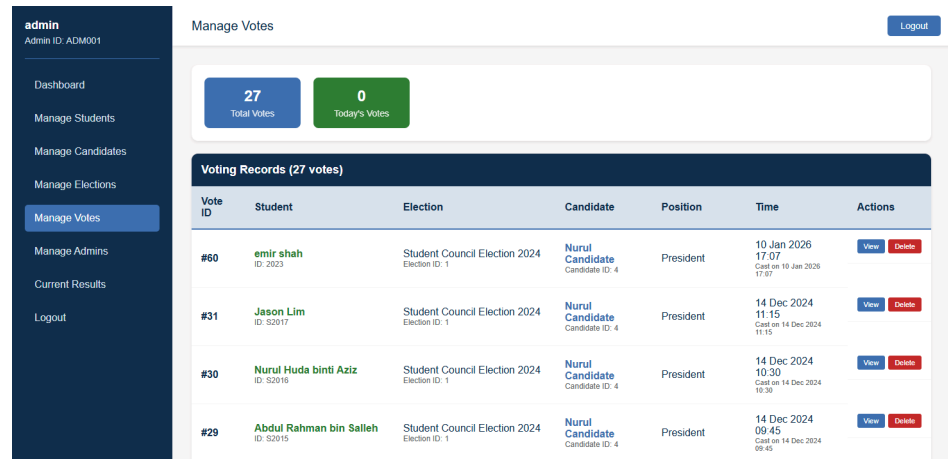


FIGURE 11 MANAGE VOTE PAGE

The Vote Management page enables administrators to monitor all voting activities across elections. This interface provides full oversight of the voting process with the following capabilities:

1. **View** : Display all voting records including vote ID, student name and ID, election title, candidate name, position, and vote timestamp in a sortable table format.
2. **Total Votes**: View system wide statistics showing total number of votes cast across all elections.
3. **Today's Votes**: Monitor real time voting activity with a dedicated counter showing votes cast on the current day.
4. **Voting Records**: Comprehensive table displaying:
 - Vote ID
 - Student information
 - Election title
 - Candidate name
 - Position contested
 - Vote timestamp (date and time)
 - Action buttons (View and Delete)
 - View Details: Show complete vote information in a popup including all vote attributes.
 - Delete Vote :Remove votes that can restore candidate vote count

4.12. MANAGE ADMIN PAGE

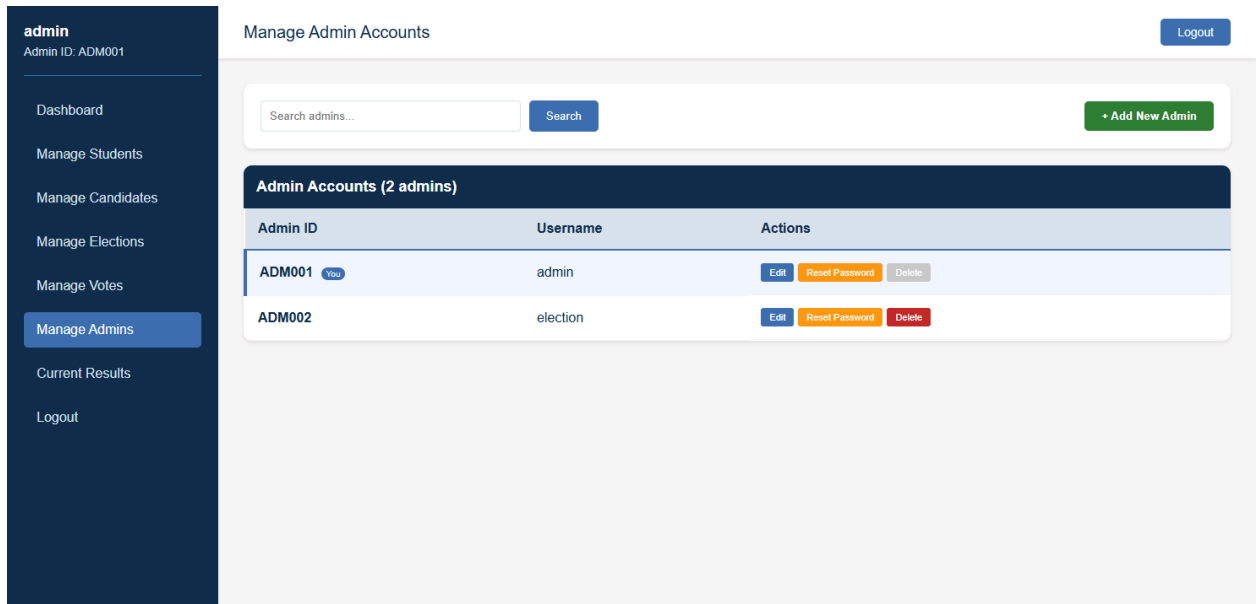


FIGURE 12 MANAGE ADMIN PAGE

The Manage Admin page enables administrators to check and manage all administrator accounts in the system. This interface provides a very good control over admin users with the following capabilities:

Core Operations:

1. **View:** Display a complete list of all admin accounts including Admin ID and Username in a sortable table format.
2. **Add:** Create new administrator accounts by providing Admin ID, username, and password with confirmation validation.
3. **Edit:** Modify existing admin usernames and optionally update passwords while maintaining account integrity.
4. **Delete:** Remove admin accounts from the system with built in protection that prevents administrators from deleting their own active account.
5. **Reset Password :** Reset admin passwords with confirmation validation and minimum length requirements
6. **Search:** Quickly filter and locate specific admin accounts using keywords, Admin ID, or username.

5. USER MANUAL

6. CONCLUSIONS

The UITM Election Management System provides a comprehensive, secure, and efficient solution for conducting campus elections. By automating the election process, the system reduces administrative workload, minimizes errors, enhances transparency, and increases student participation in the voting process i guess. The modular architecture allows for future enhancements and scalability, making it a sustainable solution for educational institutions seeking to modernize their electoral processes.